Common output bias matters in randomness certification from quantum speed limits

Xiaojian Zeng*
PBC School of Finance, Tsinghua University, Beijing 100190, China
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Randomness certification from quantum speed limits can be formulated as an entropy minimization over ensembles with a bounded average branch speed. For two equally likely preparation times and one binary measurement, a known symmetry argument gives a conservative bound by replacing the observed correlations with their center at fixed correlation difference. Whether this replacement preserves the optimum was left as a conjecture by Jones *et al.*, PRX Quantum **7**, 010349 (2026). We give a computer-assisted counterexample at dimensionless speed budget $e = 1/2$. A fixed rational affine entropy bound gives $H^\star(\sin(3/5), -\sin(2/5), 1/2) > 0.7758$ bits, whereas an exactly feasible three-branch ensemble gives $H^\star(d, -d, 1/2) < 0.7653$ bits, with $2d = \sin(3/5) + \sin(2/5)$. Both observed points are quantum admissible. The strict difference exceeds 0.0105 bits. The certificate covers the entire continuous branch domain and is reproduced with distinct interval arithmetic implementations. The result concerns the average branch-speed relaxation with classical side information; it leaves the conservative center bound valid and does not identify an optimal attack under a total-state energy-variance constraint.

I. INTRODUCTION

A bound on the speed of quantum evolution can constrain the randomness of a binary measurement performed after one of two preparation times. Jones *et al.* formulated this connection using an upper bound on energy uncertainty and classical side information [1]. Their entropy optimization retains two observed correlations, C_0 and C_1, together with an average speed budget. Its convexity and symmetries imply that replacing (C_0, C_1) by $((C_0 - C_1)/2, -(C_0 - C_1)/2)$ gives a conservative lower bound. They conjectured that this replacement always preserves the optimized entropy, immediately after Lemma 3 of Appendix E(ii) in the version-3 preprint [1]. All locations in that work below refer to arXiv version 3.

We disprove that equality conjecture with an explicit pair of correlations. The construction combines a globally valid affine entropy witness at an off-center point with a feasible ensemble at its center. Their bounds are strictly separated. No numerical estimate of either optimum is required.

The use of full binary statistics, convex-roof entropy optimization, and affine dual bounds has established precedents, including the energy-constrained analysis of Van Himbeeck and Pironio [2]. The contribution here is the strict common-bias effect for the specific average speed constraint, rather than a new optimization method. We compare the constraints in Sec. VI. This distinction also separates the result from related randomness certification using spatial symmetries [3].

* xiaojianzeng@foxmail.com

II. MODEL AND THE EQUALITY QUESTION

Consider a time-independent self-adjoint Hamiltonian, unitary evolution between preparation times separated by $\Delta t > 0$, and one binary POVM $M_\pm \geq 0$, $M_+ + M_- = I$. The input $x \in \{0, 1\}$ is uniform, and $C_x = P(+|x) - P(-|x)$. Set $e = \mathfrak{E}\Delta t/\hbar$, where $\mathfrak{E}$ is an energy-standard-deviation budget. All speed parameters below are dimensionless; calculations use $\hbar = \Delta t = 1$. No perturbative expansion or Hilbert-space truncation is made.

For a branch budget $0 \leq w \leq \pi/2$, define the correlation domain

$$Q_w = \{(u, v) \in [-1, 1]^2 : F(u, v) \geq \cos w\}, \tag{1}$$

$$F(u, v) = \frac{\sqrt{(1+u)(1+v)} + \sqrt{(1-u)(1-v)}}{2}. \tag{2}$$

This is the speed-limit domain of Eqs. (8)–(9) in Ref. [1]. Write the binary Shannon entropy as

$$h(c) = -\sum_{z=\pm 1} \frac{1 + zc}{2} \log_2 \frac{1 + zc}{2}, \tag{3}$$

$$H(u, v) = \tfrac{1}{2}[h(u) + h(v)], \tag{4}$$

with $0 \log_2 0 = 0$. Entropies are in bits per measurement round.

The precise optimization studied here is

$$\begin{aligned} H^\star(C, e) = \inf \ & \sum_\lambda p_\lambda H(C^\lambda) \\ \text{subject to } & p_\lambda \geq 0, \quad \sum_\lambda p_\lambda = 1, \\ & \sum_\lambda p_\lambda C^\lambda = C, \quad \sum_\lambda p_\lambda w_\lambda = e, \\ & 0 \leq w_\lambda \leq \pi/2, \quad C^\lambda \in Q_{w_\lambda}. \end{aligned} \tag{5}$$

It is the average branch-speed relaxation in Eq. (27) of Ref. [1], with classical hidden variable λ. There is no

imposed bound on the number of branches. The lower-bound argument below also applies if the sums are replaced by probability integrals.

The equality in the last moment constraint is a budget convention, not a requirement that every branch saturate its physical energy uncertainty. An ensemble with average label $r < e \le \pi/2$ can be padded by replacing

$$w_\lambda \longmapsto w_\lambda + \frac{e-r}{\pi/2-r}(\pi/2 - w_\lambda). \tag{6}$$

This preserves correlations and entropy, keeps every label within its allowed range, and raises the mean to e. Thus the corresponding upper-bound budget convention gives the same infimum.

Let $d = (C_0 - C_1)/2$. The previously proved center inequality is

$$H^\star(d, -d, e) \le H^\star(C_0, C_1, e). \tag{7}$$

For completeness, $H^\star$ is convex in (C, e) by mixing feasible ensembles. The branch transformations $(u, v) \mapsto (v, u)$ and $(u, v) \mapsto (-u, -v)$ preserve both Q_w and H. Hence the two points (C_0, C_1) and $(-C_1, -C_0)$ have equal optimized entropy, and their average is $(d, -d)$. This proves Eq. (7), as in the cited lemma. The question is whether equality holds for every $0 < e < \pi/2$ and $C \in Q_e$.

III. AN EXPLICIT STRICT SEPARATION

Set

$$e = \frac{1}{2}, \qquad C^{\text{off}} = (\sin(3/5), -\sin(2/5)),$$
$$d = \frac{\sin(3/5) + \sin(2/5)}{2}, \quad C^{\text{mid}} = (d, -d). \tag{8}$$

Our result is

$$H^\star(C^{\text{off}}, 1/2) > 0.7758, \tag{9}$$
$$H^\star(C^{\text{mid}}, 1/2) < 0.7653, \tag{10}$$
$$H^\star(C^{\text{off}}, 1/2) - H^\star(C^{\text{mid}}, 1/2) > 0.0105. \tag{11}$$

The first two inequalities immediately imply the third. They suffice to disprove universal equality in Eq. (7).

Both observed points lie in $Q_{1/2}$. Indeed, parameterize the entire correlation square by

$$u = \cos(2a), \quad v = \cos(2b), \qquad a, b \in [0, \pi/2]. \tag{12}$$

The corresponding probability vectors are $(\cos^2 a, \sin^2 a)$ and $(\cos^2 b, \sin^2 b)$, so

$$F(u, v) = \cos(a - b), \qquad (u, v) \in Q_w \iff |a - b| \le w. \tag{13}$$

For C^{off}, choose $a = \pi/4 - 3/10$ and $b = \pi/4 + 1/5$. Their difference is exactly $1/2$. An explicit physical realization is

$$|\psi_0\rangle = \begin{pmatrix} \cos a \\ \sin a \end{pmatrix}, \quad K = \tfrac{1}{2}\sigma_y, \quad M_+ = |0\rangle\langle 0|. \tag{14}$$

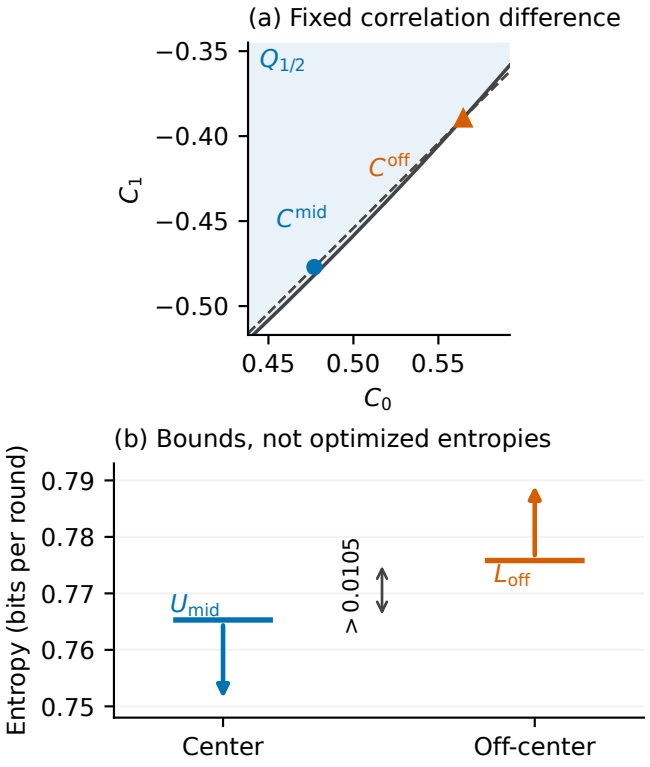


FIG. 1. Two correlations with the same difference at $e = 1/2$. (a) Shading indicates the allowed side of the $Q_{1/2}$ boundary; the dashed line has $C_0 - C_1 = 2d$. (b) Horizontal marks give the affine lower bound L_{off} and feasible upper bound U_{mid}. Arrows show the directions of the entropy inequalities; the marks are not optima. AI-assisted Python/Matplotlib code generated the plot from analytic equations and interval-certificate outputs. Floating-point geometry is for display only.

Here $\exp(-iK)|\psi_0\rangle = (\cos b, \sin b)^T$, and $\langle K\rangle = 0$, $K^2 = I/4$, so the energy standard deviation is $1/2$. Restoring units replaces K by $(\hbar/2\Delta t)\sigma_y$. The center obeys

$$d = \sin(1/2)\cos(1/10) < \sin(1/2), \tag{15}$$

and therefore has minimum branch budget $\arcsin d < 1/2$. The correlation difference is exactly $2d$ at both points. Their common biases $(C_0 + C_1)/2$ are, respectively, $[\sin(3/5) - \sin(2/5)]/2$ and zero.

IV. A GLOBAL AFFINE ENTROPY CERTIFICATE

All coefficients in the witness below are exact rationals:

$$t = -\frac{211}{50}, \quad b_0 = \frac{5907}{100}, \quad b_1 = -\frac{5487}{100}, \quad k = \frac{1989}{20}. \tag{16}$$

We establish, for every allowed branch,

$$t + b_0 u + b_1 v - k w \le H(u, v). \tag{17}$$

Averaging this inequality over any feasible ensemble gives

$$H^\star(C,e) \ge t + b_0 C_0 + b_1 C_1 - ke. \tag{18}$$

This is the affine lower-bound principle of the source's Eq. (32), but only weak duality is needed here: neither an optimizer nor a strong-duality theorem enters the proof.

Define $s(a) = h(\cos 2a)$. Because $k > 0$, it suffices to use the smallest allowed branch budget in Eq. (13). Equation (17) is therefore equivalent to $G(a,b) \ge t$ throughout $[0,\pi/2]^2$, where

$$G(a,b) = \tfrac{1}{2}[s(a)+s(b)] - b_0\cos 2a - b_1\cos 2b + k|a-b|. \tag{19}$$

If $a \ge b$, then $u \le v$ and

$$-b_0 u - b_1 v = -\tfrac{21}{5}u + \tfrac{5487}{100}(v-u) \ge -\tfrac{21}{5} > t. \tag{20}$$

Since entropy and $k|a-b|$ are nonnegative, this half of the domain is covered analytically.

If $a \le b$, the objective separates:

$$\begin{aligned} G(a,b) &= f_0(a) + f_1(b), \\ f_0(a) &= \tfrac{1}{2}s(a) - b_0\cos 2a - ka, \\ f_1(b) &= \tfrac{1}{2}s(b) - b_1\cos 2b + kb. \end{aligned} \tag{21}$$

The following finite computation, with an explicit continuum correction, covers this entire triangle.

A. Endpoint control and the continuum correction

Although the derivative of $h(c)$ diverges at $c = \pm 1$, the angular entropy has a bounded derivative. For $0 < a < \pi/2$,

$$s'(a) = \frac{4\sin a\cos a}{\ln 2}\ln(\cot a). \tag{22}$$

On $0 < a \le \pi/4$, $0 \le \ln(\cot a) \le -\ln(\sin a)$. The elementary inequality $-x\ln x \le 1/\exp(1)$ gives

$$|s'(a)| \le \frac{4}{\exp(1)\ln 2} < 3. \tag{23}$$

For the last strict bound one can use $\exp(1) > 8/3$ and $\ln 2 = \int_1^2 \mathrm{d}x/x > 1/2$. The symmetry $s(\pi/2 - a) = s(a)$ covers the other half, and the one-sided endpoint derivatives tend to zero. Thus s is globally 3-Lipschitz, including its endpoints. It follows that

$$\mathrm{Lip}(f_0) \le 219.09, \qquad \mathrm{Lip}(f_1) \le 210.69. \tag{24}$$

These terminating decimals are exact rational upper bounds.

Take $N = 32768$ and $a_j = j\pi/(2N)$, $j = 0,\dots,N$. Round a and b to the nearest nodes using the same monotone tie rule. If $a \le b$, their indices obey $i \le j$, and each distance is at most $\pi/(4N)$. Therefore

$$\begin{aligned} \min_{a\le b} G(a,b) &\ge M_N - \frac{429.78\pi}{4N}, \\ M_N &= \min_{0\le i\le j\le N}[f_0(a_i) + f_1(a_j)]. \end{aligned} \tag{25}$$

There is no unverified interpolation assumption in this estimate.

B. The finite certificate

Each node value is enclosed by directed interval arithmetic and its lower endpoint is rounded downward to a multiple of 10^{-25}. Let the resulting integers for f_0 and f_1 be A_i and B_i. Then

$$M_N \ge 10^{-25}\min_j\left[B_j + \min_{i\le j} A_i\right]. \tag{26}$$

The inner minimum is maintained as a running prefix minimum. This evaluates the bound for all ordered node pairs in linear time without omitting any pair. The full traversal gives

$$M_N \ge -\frac{420673217330444402007158 43}{10^{25}}. \tag{27}$$

Using $\pi < 22/7$, the continuum correction is bounded by

$$E_N = \frac{236379}{22937600}. \tag{28}$$

Combining this with Eqs. (25) and (27) yields

$$\begin{aligned} \min_{a\le b} G(a,b) &\ge -\frac{295192623591272018905010901}{7\cdot 10^{25}} \\ &> -4.22. \end{aligned} \tag{29}$$

Together with the analytic reverse triangle, this proves Eq. (17) over the full branch domain. The same discrete integer bound was obtained with 40-digit `mpmath.iv` intervals and a separate 48-digit Decimal implementation. A third implementation using 128-bit fixed-point integer intervals checks continuous cells directly and avoids the Lipschitz correction; see Appendix A.

Evaluating Eq. (18) at C^{off} gives

$$\begin{aligned} L_{\mathrm{off}} = &-4.22 + 59.07\sin(3/5) + 54.87\sin(2/5) \\ &- 99.45/2 > 0.7758, \end{aligned} \tag{30}$$

which proves Eq. (9).

V. AN EXACTLY FEASIBLE CENTER ENSEMBLE

Choose

$$w = \frac{66}{125}, \qquad q = \frac{d}{\sin w}, \qquad w_0 = \frac{1/2 - qw}{1-q}. \tag{31}$$

Consider the three branches in Table I. Their weights are legitimate because interval evaluation gives

$$0.94685 < q < 0.94686, \qquad 0.00116 < w_0 < 0.00118. \tag{32}$$

The anticorrelated branch has minimum budget exactly w. The other two branches are deterministic and lie in Q_{w_0} for any nonnegative w_0. Their budget labels need not be saturated by their actual energy uncertainties.

The moment constraints hold as algebraic identities:

$$q \sin w = d, \qquad qw + (1-q)w_0 = 1/2. \tag{33}$$

The two deterministic contributions cancel in the correlation mean and contribute zero entropy. Hence

$$H^\star(C^{\mathrm{mid}}, 1/2) \le U_{\mathrm{mid}} := q\, h(\sin w) < 0.7653. \tag{34}$$

This proves Eq. (10). Feasibility is exact, not equality within a linear-programming tolerance. Neither the three-branch support nor the chosen w is asserted to be optimal.

For clarity, outward-rounded interval evaluations of the two witness values and their difference give

$$\begin{aligned} 0.77581534592039102759 &< L_{\mathrm{off}} \\ &< 0.77581534592039102760, \\ 0.76529069411815580216 &< U_{\mathrm{mid}} \\ &< 0.76529069411815580217, \\ 0.01052465180223522542 &< L_{\mathrm{off}} - U_{\mathrm{mid}} \\ &< 0.01052465180223522543. \end{aligned} \tag{35}$$

These narrow intervals describe L_{off} and U_{mid}, not the two optimized entropies. The asserted separation in Eq. (11) uses only the coarser strict bounds.

VI. PHYSICAL INTERPRETATION AND SCOPE

For quantum-admissible data, replacing the off-center correlations by their center can lose more than 0.0105 bits per round in Eq. (5). The inequality in Eq. (7) remains valid: center bounds may still be propagated along lines of fixed correlation difference as conservative bounds, including the use described in Appendix E(v) of Ref. [1]. The counterexample concerns the conjectured equality, not that lower-bound procedure.

TABLE I. A feasible ensemble for (C^{mid}, e). All weights and budget parameters have the exact definitions in Eq. (31).

Probability	Correlations	Budget
q	$(\sin w, -\sin w)$	w
$(1-q)/2$	$(1,1)$	w_0
$(1-q)/2$	$(-1,-1)$	w_0

It is essential to distinguish the average branch constraint from a bound on the energy uncertainty of the total prepared state. For a fixed Hamiltonian K and $\bar\rho = \sum_\lambda p_\lambda \rho_\lambda$,

$$\begin{aligned} \mathrm{Var}_{\bar\rho}(K) = &\sum_\lambda p_\lambda \,\mathrm{Var}_{\rho_\lambda}(K) \\ &+ \mathrm{Var}_p(\mathrm{tr}\,\rho_\lambda K)\,. \end{aligned} \tag{36}$$

Consequently $\Delta K_{\bar\rho} \ge \sum_\lambda p_\lambda \Delta K_{\rho_\lambda}$. A total-state uncertainty bound therefore implies an average branch bound; the reverse implication is not automatic. Our center ensemble establishes an upper bound for Eq. (5), not an optimal physical attack under the stronger total-state constraint.

Nor can one directly identify the average speed budget with the earlier average-energy constraints of Ref. [2]. That work retains input-dependent energy expectation bounds and permits full-statistics or linear-statistic entropy optimization; see its version-1 Eqs. (26)–(29) and Remark 9. The constraints are not the scalar average of w used here. Even if an individual overlap can be matched by an energy parameter $\omega = \sin^2(w/2)$, this nonlinear change does not preserve ensemble averages. Likewise, the comparison to energy constraints in Appendix E of Ref. [3], Eqs. (55)–(58), explicitly treats a regime without a maximum-average assumption. That peak-constraint correspondence does not establish equality for the average speed problem.

The result is a single strict counterexample, with no claim of a phase diagram, exact optimized entropy, new duality method, or exhaustive priority over subsequent literature. It assumes classical side information and does not provide a quantum-side-information or finite-sample security proof. There is no spatial boundary, thermodynamic limit, perturbative order, or uncontrolled infinite-dimensional tail in the certificate. Extending the comparison to the true total-state constraint is a separate question.

VII. COMPUTATION AND REPRODUCIBILITY

The reproducibility files contain the fixed rational coefficients, the analytic ensemble definitions, and standalone Decimal and integer checkers. Both recompute the global witness and scalar bounds without external numerical packages or cached function values. The verified environment is Python 3.9.6; the Decimal backend is version 1.70 with libmpdec 2.5.0. The integer checker uses only Python integer and rational arithmetic. Computational correctness relies on those arithmetic implementations and the explicit error arguments, and has not been formalized in a proof assistant.

AI assistance was substantive: OpenAI Codex was used for literature comparison, counterexample search, derivations, certificate code, numerical checks, and interpretation. The writing environment identifies its model

as GPT-6; exact model revisions across the earlier research sessions have not been independently established. Separately implemented arithmetic checks provide the computational cross-checks described above. They do not constitute external peer review. Human scientific review and approval of this local draft remain pending.

ACKNOWLEDGMENTS

OpenAI Codex, identified as GPT-6 in the writing environment, also assisted with drafting and revising the manuscript, bibliography preparation, and figure code. Its research contributions and verification limits are described in Sec. VII.

DATA AVAILABILITY

All data needed to reproduce the stated bounds are specified in the paper and the accompanying local reproducibility files. These include two standalone checkers, certificate outputs, a figure script, and instructions. No external experimental dataset is used. A public archive location has not yet been assigned.

Appendix A: Arithmetic and an independent continuous-cell bound

1. Directed Decimal node evaluation

The Decimal checker uses 48-digit outward rounding for addition, multiplication, and division. It encloses π through

$$\pi = 16\arctan(1/5) - 4\arctan(1/239), \tag{A1}$$

using 40 terms of each rational alternating series and the next-term remainder. Cosine is evaluated through degree 58 and sine through degree 59, with absolute remainder bounds $(22/7)^{60}/60!$ and $(22/7)^{61}/61!$, respectively. The logarithm of each interval endpoint is evaluated by Decimal's correctly rounded nearest logarithm [4], then widened to its neighboring representable numbers. Monotonicity encloses the logarithm throughout the input interval. Probabilities are reconstructed as $(1 \pm \cos 2a)/2$; at the two exact endpoints the entropy is set to zero. This gives all A_i, B_i of Eq. (26) by exact conversion of decimal endpoints to rationals followed by downward integer rounding.

2. Direct interval-cell coverage

An independently implemented check avoids the nearest-node and Lipschitz arguments altogether. Partition $[0, \pi/2]$ into $n = 65536$ cells $I_j = [\alpha_j, \alpha_{j+1}]$, with $\alpha_j = j\pi/(2n)$. The entropy s increases up to $\pi/4$ and decreases afterwards. Set

$$m_j = \min\{s(\alpha_j), s(\alpha_{j+1})\}.$$

This smaller endpoint value bounds s below throughout the cell, including a cell containing the maximum. Cosine decreases over the angular range. It follows that, for every $\theta \in I_j$,

$$\begin{aligned} f_0(\theta) &\geq \tfrac{1}{2}m_j - b_0 \cos 2\alpha_j - k\alpha_{j+1} =: \mathcal{A}_j, \\ f_1(\theta) &\geq \tfrac{1}{2}m_j - b_1 \cos 2\alpha_{j+1} + k\alpha_j =: \mathcal{B}_j. \end{aligned} \tag{A2}$$

Choosing cell indices monotonically, with $\pi/2$ assigned to the last cell, ensures $i \leq j$ whenever $a \leq b$. Therefore

$$\min_{a \leq b} G(a,b) \geq \min_j \left[\mathcal{B}_j + \min_{i \leq j} \mathcal{A}_i\right]. \tag{A3}$$

The same-cell rectangle may include points outside the triangle, but this only lowers the bound. The prefix algorithm again covers every ordered pair, using $O(n)$ work and constant auxiliary storage.

This implementation represents interval endpoints as integers divided by $S = 2^{128}$, rounding outward after every operation. It encloses π by $4[\arctan(1/2) + \arctan(1/3)]$ with 80 rational alternating terms in each arctangent. Cosine uses the Taylor polynomial through degree 48 and remainder $(22/7)^{50}/50!$. For logarithms, powers of two reduce a positive argument to $y \geq 1$; the code verifies $0 \leq z = (y-1)/(y+1) \leq 1/2$ and uses

$$\ln y = 2\sum_{j=0}^{55} \frac{z^{2j+1}}{2j+1} + R, \qquad 0 \leq R \leq \frac{8}{3 \cdot 113 \cdot 2^{113}}. \tag{A4}$$

The same series encloses $\ln 2$ and reconstructs the original logarithm. This backend calls no floating-point trigonometric or logarithmic function.

The complete traversal gives $\min_{a \leq b} G(a,b) \geq R_{\mathrm{cell}}$, where the conservative bound is the exact rational number

$$R_{\mathrm{cell}} = -\frac{143309901745618643428259319073050687372 7}{2^{128}}. \tag{A5}$$

It is greater than -4.211500 and hence greater than $t = -4.22$. This independently certifies the unchanged witness. The same integer arithmetic yields Eq. (35). No part of the conclusion depends on agreement of unbounded-precision decimal approximations alone.

[1] C. L. Jones, A. Aloy, G. Higgins, and M. P. Müller, Certified randomness from quantum speed limits, PRX Quantum **7**, 010349 (2026), equation and appendix locators in the text refer to arXiv:2506.14526v3, arXiv:2506.14526 [quant-ph].

[2] T. Van Himbeeck and S. Pironio, Correlations and randomness generation based on energy constraints (2019), version 1, arXiv:1905.09117 [quant-ph].

[3] C. L. Jones, S. L. Ludescher, A. Aloy, and M. P. Müller, Theory-independent randomness generation from spatial symmetries, Quantum **10**, 1966 (2026), appendix locators refer to arXiv:2210.14811v4, arXiv:2210.14811 [quant-ph].

[4] Python Software Foundation, Python 3.9 documentation: decimal—Decimal fixed point and floating point arithmetic (n.d.), entries Decimal.ln, Context.ln, next_minus, and next_plus; accessed September 12, 2026.